Twin-Satellite Constellation Design and Realization for Terrain Mapping and Deformation Monitoring: LuTan-1

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***Abstract*— LuTan-1 (LT-1) is the first civil L-band synthetic aperture radar (SAR) satellite constellation and comprises two identical satellites. LT-1 is designed to fulfill two main requirements, one of which the main tasks is to provide digital surface model (DSM) products covering the areas that are not available using optical satellites. The second task is to provide deformation products to support the geohazard monitoring task. For the first task, LT-1 provides a novel noninterrupted imaging technology to facilitate phase synchronization. For the second task, the orbit maintenance is implemented using a newly proposed in-plane offset semimajor axis and out-of-plane control triggered strategy. Besides, we provide the system performance analysis for the two main tasks. Finally, the first results are produced with root-mean-square error (RMSE) of the DSM less than 0.7 m in the flat region and 6.7 m in the mountainous region. The RMSE of the first deformation products is less than 2.7 mm in the test region. The results indicate a favorable potential for terrain mapping and deformation monitoring applications.**

***Index Terms*— Deformation, digital surface model (DSM), interferometric synthetic aperture radar (InSAR), LuTan-1 (LT-1).**

I. Introduction

TWIN-SATELLITE constellation is commonly used in interferometric synthetic aperture radar (InSAR) applications such as digital surface model (DSM) generation, forest monitoring, and deformation monitoring. Although there have been ERS-1/2 satellite constellations since 1991, they did not communicate with each other after launch [1]. The first twin satellites were TanDEM-X which were especially designed for DSM generation [2]. Phase synchronization was first applied in synthetic aperture radar (SAR) satellites using synchronization horns onboard the satellites. LuTan-1 (LT-1) is the second twin-satellite constellation with the mission of DSM generation and deformation monitoring. TanDEM-L is a well-elaborated and published mission proposal that allows for interferograms worldwide to observe the dynamics of the Earth twice every week [3]. Harmony missions were proposed to observe the world's minor changes using two C-band SAR instruments that can only receive signals from Sentinel-1 satellites [4]. The main purpose of the twin-satellite constellation is to ensure that the sending or receiving conditions are identical between the satellites, thus providing interferograms with less temporal coherence loss.

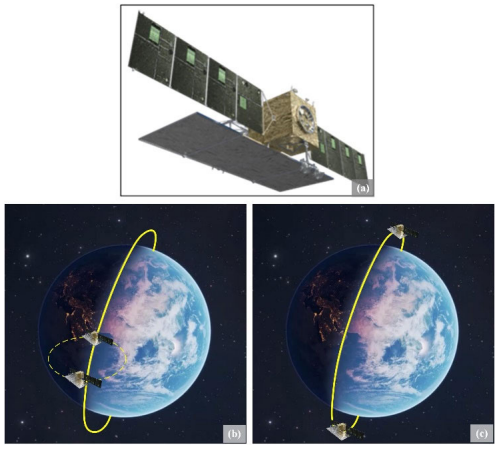


Fig. 1. Formations of LT-1. (a) Architecture of the satellite, (b) HBF used for DSM generation, and (c) PMF used for deformation monitoring.

LT-1 is the first SAR satellite constellation used for deformation monitoring in China. LT-1 comprises two identical satellites, i.e., LT-1A and LT-1B. These two satellites were designed following the same parameters. LT-1 supports two formations (as shown in Fig. 1). The first is the TanDEM-X-like helix bistatic formation (HBF) [2], in which the satellites fly around each other to provide DSM products. The second is pursuit monostatic formation (PMF), in which the satellites follow each other to provide deformation products. These two formations were designed to fulfill the needs of DSMs in cloudy and rainy regions of China and quantitative deformation observations over highly and moderately susceptible geohazard areas (HIMSGAs) in China.

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Fig. 2. Difficult SAM regions of optical satellites.

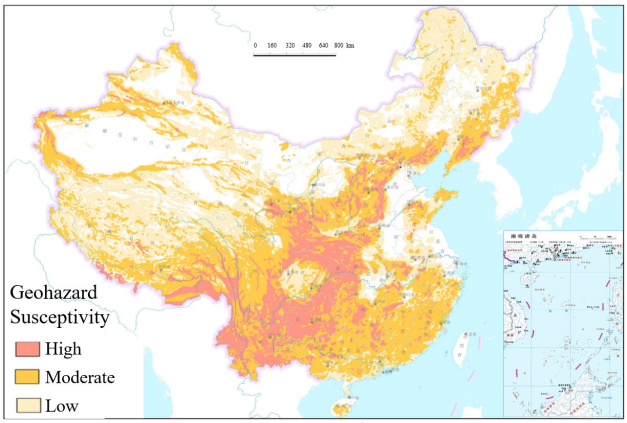


Fig. 3. HIMSGAs in China.

A. Need for DSM Data

The shape of the Earth's surface constitutes the basic information of geoscience. Spaceflight requires DSM data to provide a more precise gravity model for orbit prediction and control. National economic construction is supported by elevation data, especially digital line graphs, which encompass building locations generated using orthorectified images and contour lines obtained via DSMs. The DSM represents the geometric explanation of the lithosphere and hydrosphere, reflecting the atmospheric structure in the hydrosphere and the biological distribution of the biosphere. In China, surveying and mapping (SAM) tasks have been conducted using photogrammetry supported by the Ziyuan-3 and Gaofen-7 optical payloads. However, these optical payloads cannot collect data covering the cloudy and rainy regions located in southwestern China. After years of data collection and processing, an area of approximately 330 000 km^2 remains covering eastern Sichuan, western Guizhou, southern Yunnan, and southern Tibet, which cannot be reached by optical satellites (refer to Fig. 2).

InSAR is more efficient than optical photogrammetry when calculating the elevation of the Earth's surface [5]. In the Shuttle Radar Topography Mission (SRTM), data were collected for 11 days, i.e., from 56°S to 60°N, from the DSM of the Earth's surface using the C-band. TanDEM-X is the second SAM SAR mission providing a worldwide X-band DSM covering even the North Pole [2]. The China SAM SAR satellite plan was initiated more than ten years ago, aiming at providing DSM data with a unified reference, unified data source, and unified accuracy. LT-1 demonstrates the operational usage of InSAR technology for SAM missions in China.

B. Need for Deformation Data

The Earth's surface is persistently deformed in response to geohazards, including landslides, volcanoes, and debris flows. China contains various terrain types and nearly all geohazard types. The HIMSGA in China covers a total area of approximately 3 940 000 km^2 (please refer to Fig. 3). Most of the areas that are highly susceptible to geohazards occur in the second-level terrain ladder consisting of the Sichuan–Yunnan Mountainous Regions, Yunnan–Guizhou Highland, Qinling–Bashan Mountainous Regions, and Middle and Southern Gansu. The areas that are moderately susceptible to geohazards are distributed in Eastern Tibet, Northwest Mountainous Regions, Central China, Southeast Hilly Regions, and Northeast China. Early warning is necessary to prevent sudden geological disasters in the HIMSGA. InSAR technology is the only way to provide continuous slow deformation observations as small as centimeters or millimeters in size in space. LT-1 can provide data with orbit repeat days as short as four days over the areas of interest.

As the first InSAR satellite constellation, LT-1 solved mainly three kinds of problems. The first problem is satellite manufacturing to ensure satellite performance, including two difficulties and innovations such as how to find and maintain the interferometric orbit and how to select the phase synchronization devices and implement it. The second problem is satellite parameters' demonstration and performance analysis for DSM and deformation applications. The height accuracy should fulfill the needs of 1:50 000 measurement scale. The final differential InSAR (DInSAR) accuracy is 5 cm. The interferometric stochastic models have been established for error propagation and performance analysis of those two main application tasks. The third problem is the operational application of the data in China. Key techniques of the three parts are discussed in this article.

The outline of this article is as follows. Section II introduces the satellite system, which includes orbital maintenance, orbital determination, and payload features. Section III provides the system performance for DSM generation. Section IV describes the system performance for deformation monitoring. We finally provide the DSM and deformation products in Section V. Conclusions are drawn in Section VI with a short summary and outlook on present and future work.

II. Satellite System

A. Mission Overview

LT-1 uses the L-band system, and the wavelength is approximately 23.5 cm, with a focus on maintaining a high interferometric coherence. The payload is fully polarized. The repeat orbit of LT-1 is eight days for a single satellite and four days for two satellites in PMF. The perpendicular baseline of the HBF ranges from 700 to 7000 m to ensure that the

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TABLE I
TERRAIN TYPES AND DEFINITIONS

Terrain type	Slope (°)	Elevation deviation (m)
Flat ground	<2	<80
Hills	[2, 6)	[80, 300)
Mountainous land	[6, 25)	[300, 600)
Alpine region	≥25	≥600

height of ambiguity (HoA) ranges from 7 to 160 m. The DSM accuracy is defined at a 1:50 000 measurement scale. The grid size is 25 m. The accuracy of the DSM is determined by different terrain types (refer to Table I). The elevation root-mean-square error (RMSE) of the four main terrain types should be less than 3, 5, 8, and 14 m [6]. The orbital tube radius is 350 m, indicating that the maximum perpendicular baseline is less than 700 m in the PMF. The baseline coherence can be ensured to exceed 0.95 in this formation. LT-1 uses the X-band to transfer data, and the data transfer rate is 2 × 450 Mb/s, which allows us to collect the data all over the HIMSGA every 28 days using Stripmap 1 and 20 days using Stripmap 2.

LT-1 supports six imaging modes. Stripmaps 1 and 2 are the operational modes. Stripmap 1 is especially designed for DSM generation. The synchronization between LT-1A and LT-1B is considered only under stripmap 1. Stripmap 2 is specifically designed for geohazard monitoring. The swath width of stripmap 2 is approximately 100 km, thus ensuring more integrated coverage of the HIMSGA in China. Stripmaps 3 and 4 are useful for forest monitoring, emergency management, and land usage classification. Stripmap 5 can also be used in deformation monitoring, but it can only be used for experiments. The swath width of the scanning SAR (ScanSAR) mode is 400 km, which is advantageous for ocean monitoring. Full polarization is supported by the satellites. However, Stripmaps 1, 2, and 5 and ScanSAR can only use horizontal–horizontal (HH) or vertical–vertical (VV) polarization. Stripmap 3 supports HH + HV and VV + VH. Only Stripmap 4 supports full polarization.

B. SRO Design and Maintenance

The satellite strict regression orbit (SRO) was designed to fulfill the needs of an orbital tube of 350 m. The orbit was designed by applying layer-by-layer iterative formation [2]. Then we refine the initial SRO by adopting the deviation of the initial and final states as the constraint condition and via iterative calculation, an artificial reference orbit that meets the strict regression requirements can be obtained (refer to Fig. 4). This is important for satellite orbit maintenance and we finally designed the orbit the radius of which is only several centimeters.

1) Maintenance of the Single-Satellite SRO: We use a 70th-order model of Earth's gravity field to design the SRO. However, the orbit of the satellite is affected not only by the Earth's gravitational field but also by atmospheric resistance, the Sun–Moon three-body gravitational force, solar light pressure, etc. The influence of the satellite orbit deviates from the reference orbit because the attenuation of the semimajor axis under the action of atmospheric drag varies with the latitude

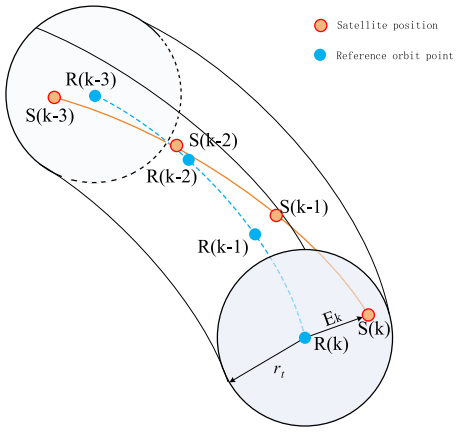


Fig. 4. Diagram of a satellite flying along an SRO.

of the Earth, and the deviation is the smallest near the Earth's poles and the greatest near the Earth's equator. The simulated value is about 15 m/day if we use the parameters of the high solar activity year, leading to about 200 m/day for the orbital tube deviation. The value is accumulated quickly. We need to conduct periodically maintenance control to decrease the effect. The three-body gravity changes the orbital inclination and the ascending node periodically in two aspects. The first is the 14-day periodical short-wave effect, and the second is the half-year periodical long-wave effect. Although the orbital tube deviation is larger than 2000 m, only the constant control is needed when it exceeds the orbital error threshold. The solar light pressure, the solid tide, and the relativity effect are small. They make the semimajor axis error accumulated by time, the argument of latitude error accumulated slowly, and inclination and ascending node change periodically. If the orbital error introduced by atmospheric and three-body gravity can be controlled, the solar light pressure can be ignored.

The parameter trajectory state due to orbital out-of-plane differences is mainly determined by the longitudinal ascending node bias at low latitudes and mainly determined by the orbital inclination bias at high latitudes. Therefore, we must compensate for the orbital perturbation introduced by the non-Earth gravitational field to maintain the SRO. Moreover, we adopted the in-plane offset semimajor axis and out-of-plane control triggered strategy when the inclination angle exceeded the threshold value, aiming at reducing the cost of fuel consumption and satellite control frequency. The main effects we have to consider is the atmospheric drag, leading to the semimajor axis decrease. We dynamically biased the semimajor axis to achieve successive overshoot in-plane control for LT-1. Then the orbit decreases due to natural perturbation. But before that, we first bias the latitude argument to decrease the longitudinal ascending node, i.e.,

$$\Delta u = \frac{n \cdot \Delta\Omega \cdot \sin i}{\omega_E} + (E_{K\max} - a \cdot \Delta i \cdot \sin u)\frac{n}{\omega_E} \quad (1)$$

where u and Δu are the value and bias of latitude argument, respectively, n is the reference orbital angular velocity, $\Delta\Omega$ is the change in the longitudinal ascending node, i and Δi are the value and bias of the inclination angle, respectively, ω_E is the rotational angular velocity of the Earth, $E_{K\max}$ is

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the bias vector of the orbit, and a is the semimajor axis. The first item on the right side is used to eliminate the bias of the longitudinal ascending node, while the second is used to overshoot the orbit by considering the maximum orbital error and the semimajor axis attenuation. If the real bias of the latitude argument is greater than Δu, we overshoot the semimajor axis using

$$\Delta a = \frac{4a(\Delta u' - \Delta u)}{3nt} \tag{2}$$

where Δa is the bias of the change in the semimajor axis, $\Delta u'$ is the real bias of the latitude argument, and t is the temporal baseline length.

Due to the control residuals and the influence of the gravity of the Sun and Moon, the orbital inclination changes unidirectionally within a limited boundary over a certain period. Unidirectional change gradually compresses the semimajor axis offset and tangential offset of successive overshoots to reduce the control frequency. When the out-of-plane parameters of the LT-1 satellites exceed a specific threshold, out-of-plane orbit control is triggered. The longitudinal ascending node may introduce ten times more orbital error compared with inclination. However, we can eliminate the effect by biasing the latitude argument. It is not necessary to control this factor. During the in-orbit test phase, orbit maintenance was performed every week by injecting fuel.

2) Maintenance of the Dual-Satellite SRO: The new features of dual-satellite SRO maintenance relative to single-satellite maintenance mainly include the constraints of dual-satellite orbiting formation on orbit maintenance, i.e., configuration accuracy and intersatellite safety. To ensure intersatellite safety and reduce the influence of SRO maintenance on formation configuration coupling, we adopted the same orbital maintenance strategy for both the satellites at the same time during the PMF phase. The speed increment and jet time are the same and are determined by the state of the chief satellite, but the jet duration differs and can be calculated according to the actual mass and thrust of each satellite.

C. Orbital Determination

1) Baseline Measurement Instruments: Both the LT-1A and LT-1B satellites are equipped with identical GNSS receivers (refer to Fig. 5). The GNSS receiver largely comprises a power supply, frequency reference unit, low-noise processing unit, GNSS information processing system, and interface connection functional modules. The GNSS receivers support GPS and Bei Dou-2 (BD2) and Bei Dou-3 (BD3) system signals from the antenna. In addition to GPS L1 and L2 and BD B1I and B3I signals, we supported the tracking of new signals B1C and B2A from the BD system. After low-noise amplification processing and dual-GNSS system splitting, the GNSS receiver completes filtering, amplification, automatic gain control, downconversion, and analog-to-digital conversion sampling. At the same time, the digital signals are transmitted to the digital signal processing part. The digital signal processor (DSP) obtains the original observation data via an acquisition, tracking, and demodulation process. It also completes navigation, positioning, and absolute timing.

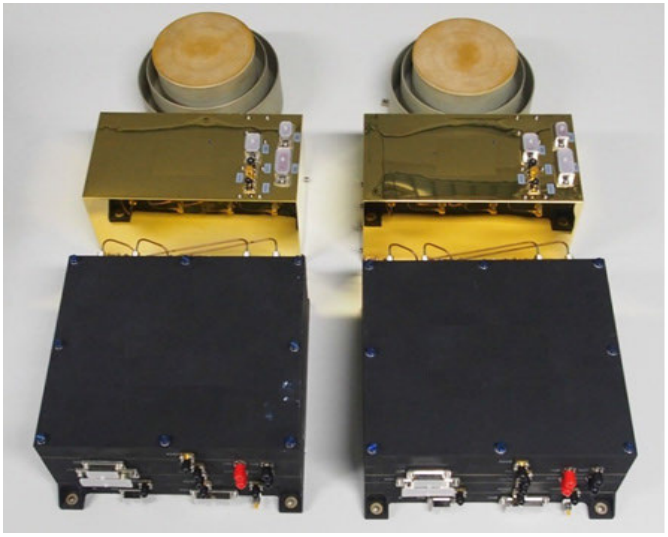


Fig. 5. GNSS receiver carried by LT-1.

The original observation data and navigation and positioning results are output to the integrated electronic subsystem and measurement and control subsystem. Moreover, the synchronization time pulse adjusted by the absolute timing is output to the integrated electronic subsystem, attitude, and orbit control subsystem and SAR subsystem.

2) Relative Navigation Implementation: The collected pseudorange and carrier phase observations onboard the LT-1 satellites are mainly used for absolute and relative orbit determination, respectively.

The satellite distance between LT-1A and LT-1B in the HBF phase is around 2 km. With respect to this baseline distance and based on a high-quality multifrequency GNSS receiver, precise baseline determination for formation-flying satellites can usually be achieved with the carrier phase differential GNSS (CDGNSS) technique, which can eliminate the influence of common errors such as atmospheric effects and thus achieve a millimeter-level relative positioning accuracy [7]. In addition to the CDGNSS technique, the reduced dynamic method can be used for relative orbit determination. A complete flowchart is shown in Fig. 6.

First, GNSS data preprocessing constitutes a prerequisite for high-precision baseline determination, which mainly includes outlier removal and cycle slip detection. Second, the zero-difference (ZD) reduced dynamic method was adopted for precise absolute orbit determination of both the formation-flying satellites, and the results served as the prior orbit for dual-satellite relative orbit determination. Then, double-difference (DD) GNSS observations were obtained by subtracting between-receiver differences, between-satellite differences, receiver and GNSS satellite clock offsets, and ephemerides from the GNSS measurements, and other common errors were eliminated and largely mitigated. Moreover, the influence of the ionosphere delay was removed by the ionosphere-free combination of DD observations, which serves as the basic observation model for relative orbit determination. Finally, one of the formation-flying satellites was selected as the reference satellite, its trajectories were fixed, and the

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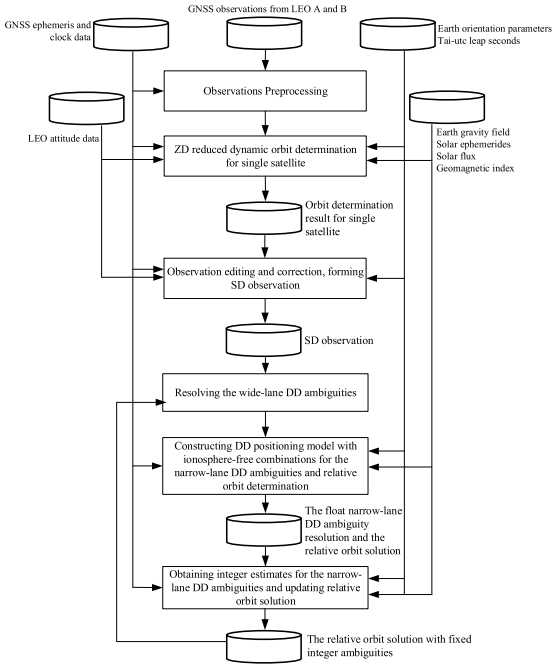


Fig. 6. Intersatellite baseline determination processing for the formation-flying satellites.

precise relative states could be obtained using the reduced dynamic method.

D. Payload Features

The payload of LT-1 features a high-performance active phased array antenna and a multimode signal generator. The antenna is split into two halves along the along-track direction that allows different polarizations to be received at the same time or for imaging with a high resolution and wide swath. The signal generator can generate a general linear frequency-modulated (LFM) chirp signal.

1) Antenna Subsystem: A large antenna is essential for obtaining radar images with a wide swath and high spatial sensitivity. Exploiting flexible beam pointing control and precise beamforming, active phased array antennas are deployed on modern spaceborne SARs such as TerraSAR-X/TanDEM-X, Advanced Land Observation Satellite-2 (ALOS-2), and GaoFen-3. In the LT-1 mission, each satellite is equipped with a complex phased array antenna. The high-resolution wide-swath (HRWS) imaging mode is indispensable for Earth observation. Azimuth multichannel technology can overcome the inherent limitations of the single aperture systems by decreasing the pulse repetition frequency (PRF) to achieve a high resolution over a wide swath. The HRWS mode of LT-1 uses dual channels to achieve a 50-km swath and 3-m resolution. Specifically, LT-1 deploys a microstrip planar array antenna designed to operate at 1.26 GHz with a bandwidth of 80 MHz and a size of 9.80 m (azimuth) × 3.4 m (elevation), as shown in Fig. 7. The antenna is capable of full polarization operation. As shown in Fig. 7, the antenna comprises four panels. Each panel has four modules, each module encompasses 22 subarrays along the elevation direction, and each subarray comprises four patches along the azimuth direction that operate as a single radiating element. In addition, the

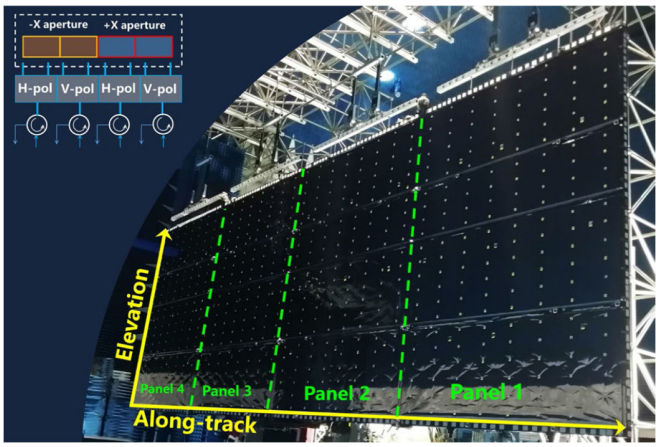


Fig. 7. Active phased array antenna of LT-1 in a microwave anechoic chamber. The antenna comprises four panels and is 9.8 m along the along-track direction and 3.4 m along the elevation direction. The subimage shown in the upper left is a diagram of the two independent apertures of the antenna [8].

four panels of the antenna can be grouped into left and right apertures, and the signal reception of each aperture is independent. The antenna radiation efficiency is greater than 80%, and the weight is approximately 1100 kg. The antenna is lightweight relative to its large size.

2) Noninterrupted Phase Synchronization Scheme: In the bistatic SAR system, different local oscillators onboard each platform are used for modulation and demodulation. The frequency deviation between the oscillators causes a residual demodulation phase over the recorded signal of the receiving-only satellite. The residual phase error can cause a nonnegligible error in the interferometric phase, which in turn reduces the height measurement accuracy. To fulfill the stringent requirements for interferometric phase stability, it is necessary to measure and compensate for the deviation between individual oscillators.

In the LT-1 mission, interferometric data acquisition is performed in the bistatic stripmap mode with two beams in the azimuth. When the interrupted synchronization method was applied, the missing echo effect and the undersampling effect were coupled to further degrade the imaging performance. Therefore, an advanced noninterrupt synchronization scheme is proposed for the LT-1 mission. We overview the method as follows [9].

A hardware structure diagram of the LT-1 system is shown in Fig. 8. The system comprises a signal generation unit (orange rectangular blocks), transmit–receive unit (yellow rectangular blocks), and calibration unit (green rectangular blocks). The signal generation unit, which comprises the GNSS disciplining module, reference frequency source, and frequency modulation signal source, is arranged to generate the linear frequency modulation signal for imaging or synchronization. The GNSS provides a time–frequency signal for the reference frequency source module, and multiple working frequency signals, which are used for the generation of the corresponding radar signal by the frequency modulation signal source, are produced by the reference frequency source. The transmit–receive unit, which consists of the microwave combination, synchronization transceiver, receiver, and data former, controls the transmission and receiving of the radar

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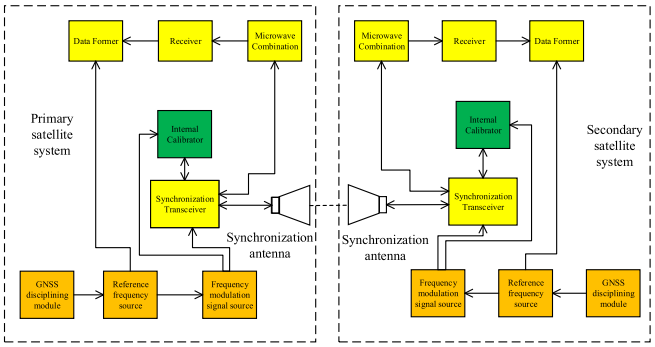


Fig. 8. Signal transmission diagram for the payload [9].

signal. In the transmitting mode, the synchronization signal is transmitted through the synchronization transceiver and synchronization antenna. In the receiving mode, the synchronization signal is received by the synchronization antenna, and then, the signal is routed through the synchronization transceiver and microwave combination. Finally, the synchronization signal is demodulated and recorded by the receiver, and after processing by the data former, the synchronization signal is then sent to the ground system.

The system operates on a separate channel with the communicating antennas onboard the satellites. We use the L-band which is the same as the carrier frequency. The purpose is to make the system easier to design. However, we avoid interference with the SAR antenna because we send the synchronization signal in the free duration between the transmitting and receiving signals. We exchange the synchronization pulse using two free durations. The first duration is between the ending time of transmitting the radar signal and the staring time of radar echo receiving window. The second duration is between the ending time of the radar echo receiving window and the starting time of the next pulse repetition time. We select the longer one from the durations to transmit signals. The pulse repetition interval of Stripmap 1 is 580 μs, the echo receiving window is 250 μs, and there are enough spare time for the 110-μs SAR pulsewidth. The maximum frequency of the synchronization can be set to 1/2 PRF. Using this method, we provide phase synchronization information with errors smaller than 2.8° [9].

III. System Performance for DSM Generation

LT-1 was designed for two main purposes, i.e., elevation and deformation monitoring. The elevation accuracy fulfills the requirements of a 1:50 000 measurement scale, and the deformation accuracy reaches 5 cm using DInSAR technology and 10 mm/year using multitemporal InSAR (MTInSAR) technology. We did not discuss new application algorithms. In this section, we conducted error propagation and performance analysis for InSAR. In section IV we did this for DInSAR and MTInSAR. For the first time, we provide the spatial error distribution model for the parallel baseline. And we have first modeled the phase unwrapping error (PUE) versus different HoA values. The main error sources and characteristics are analyzed below.

The two satellites use HBF for DSM generation, as shown in Fig. 9. They fly around each other and communicate using dedicated horn antennas. Three synchronization parameters

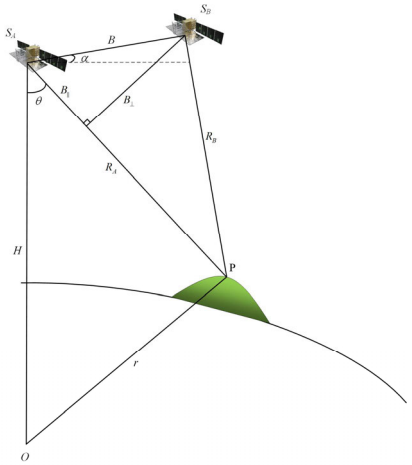


Fig. 9. Imaging geometry of LT-1 for interferometric elevation calculations.

meet the requirement of DSM generation requirements. The spatial synchronization is greater than 90%. The temporal synchronization is less than 100 ns. The phase synchronization is less than 2.8° [9]. The interferometric phase can be converted into height as follows:

$$\begin{aligned} h &= \left|\overrightarrow{\mathrm{OP}}_{XYZ}\right| - r \\ &= \left|\overrightarrow{\mathrm{OS}_A}_{XYZ} + \overrightarrow{S_AP}_{XYZ}\right| - r \\ &= \left|\overrightarrow{\mathrm{OS}_A}_{XYZ} + \boldsymbol{T}_{\mathrm{TLP}} \cdot \overrightarrow{S_AP}_{\mathrm{TLP}}\right| - r \\ &= \Bigg|\overrightarrow{\mathrm{OS}_A}_{XYZ} \\ &\quad + R_A \cdot \begin{bmatrix} \dfrac{\vec{v}}{|\vec{v}|} & \dfrac{(\vec{v}\otimes\vec{B})\otimes\vec{v}}{|(\vec{v}\otimes\vec{B})\otimes\vec{v}|} & \dfrac{(\vec{v}\otimes\vec{B})}{|(\vec{v}\otimes\vec{B})|} \end{bmatrix} \\ &\quad \cdot \begin{bmatrix} \dfrac{\lambda f_{\mathrm{dop}}}{2|\vec{v}|} \\ \dfrac{R_A}{2\left|\overrightarrow{B}_{pv}\right|}\left[1 + \dfrac{|\vec{B}|^2}{R_A^2} - \left(1 - \dfrac{\lambda\varphi}{2\pi R_A}\right)^2\right] - \dfrac{b_v r_v}{\left|\overrightarrow{B}_{pv}\right|} \\ \pm\sqrt{1 - \mathbf{v}_e^2 - \mathbf{l}_e^2} \end{bmatrix} \Bigg| - r \end{aligned} \tag{3}$$

where O is the Earth center, S_A is the phase array center (PAC) of the LT-1A satellite, S_B is the PAC of the LT-1B satellite, P is assumed to be an arbitrary point, $\overrightarrow{\bullet}_{XYZ}$ indicates that the vector can be expressed using the XYZ-axis of the 1984 World Geodetic System (WGS84), $\overrightarrow{\bullet}_{\mathrm{TLP}}$ denotes the vectors expressed using the velocity-line-of-sight-perpendicular (VLP) basis, $\boldsymbol{T}_{\mathrm{TLP}}$ is the transformation matrix from the VLP system to WGS84, $\overrightarrow{S_AP}$ is the line-of-sight (LOS) direction, r is the Earth radius below point P, $\vec{v}$ is the satellite velocity, $\vec{B}$ is the baseline, f_{dop} is the Doppler frequency, and ϕ is the unwrapped phase.

A. Positioning Accuracy of the LT-1 APC

We used the GNSS to provide accurate APC positioning information. The GNSS comprises three navigation systems: the global positioning system (GPS), BD2, and BD3. These three systems ensure that the accuracy of the APC after post-processing using precise ephemeris data is better than 10 cm. The error of the APC position is geometrically propagated to

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the location of point P. The final DSM positioning error is expected to be less than 25 m at the 1:50 000 measurement scale. Therefore, the location error of the GPS data should not be considered the main positioning accuracy error source of the LT-1 APC.

The azimuth location error is determined by the clock misalignment between the platform and payload, satellite dynamic error, orbital error, ground feature error, and geodynamics error [10]. Among these five main location error sources, clock misalignment is the main error source and remains stable over a long period, i.e., 3–6 months. This error could be larger than 2 ms, which could induce an azimuth location error of approximately 15 m given that the satellite velocity is approximately 7.5 km/s. The satellite dynamic error is introduced by the movement between the signal transmission and receiving times. This error is neglected when using the stop-and-go model, which uses the APC position considering both the signal transmitting and receiving times. The ground features are related to the main reflector within a single resolution. If a persistent scatterer (PS) exists, the pixel location accuracy increases to 1/1000, while the occurrence of a distributed scatterer (DS) decreases the pixel location accuracy to approximately 1 pixel or worse. Geodynamic errors mainly result from the Earth tides affected by surface changes induced by the gravity of the Sun and Moon. Under the influence of Earth tides, the radius increases by more than 40 cm in the radial direction, but this increase remains at the centimeter level in the horizontal direction. In addition, geodynamic errors include the Earth rotation error, which can be estimated using the International Earth Rotation Service (IERS); the Earth polar tide induced by the Earth pole change, which is less than 25 mm along the radial direction; ocean-movement-related displacement, which is less than 1 cm in inland areas; and atmospheric and hydrological loads, which normally affect the final location result at the millimeter level [11].

The slant range error sources are similar to those of the azimuth errors except for the timer inaccuracy and atmospheric delay. The SAR system can be used as an accurate timer in InSAR-related applications. The slant range can be calculated using the difference between the signal transmitting and receiving times. There is a small timing error induced by the time delay between signal generation and arrival at the antenna. Usually, this error is approximately 100 ns, leading to a slant range error of approximately 30 m before calibration. The atmospheric delay is caused by frequency-related factors, such as the ionospheric delay, and the frequency-free factors, such as the tropospheric delay. The ionospheric delay is often described by the total electron content unit (TECU), which commonly varies between 5 and 10. If we adopt 5 TECUs as an example, the ionospheric delay could reach 1.29 m along the zenith angle direction for LT-1. The tropospheric delay is approximately 3 m for the L-band. We used the generic atmospheric correction online service (GACOS) [12], [13] to mitigate atmospheric contamination.

We removed random location error sources using external data such as the IERS and GACOS data by oversampling the data to meet the resolution of multilooked single-look complex (SLC) images. Corner reflectors (CRs) were used to suppress ground feature errors (refer to Fig. 10). The random error after calibration should contribute to a location error of approximately 2 m. If we consider Earth surface elevation data, the geolocation accuracy could be better than 25 m for the final products at the 1:50 000 measurement scale.



Fig. 10. CR used to conduct the geometric calibration.

TABLE II
POSITIONING ERROR OF LT-1

Error sources	Typical value	Error before calibration	Error after calibration
APC positioning error	0.01 m	0.01 m	0.01 m
Clock misalignment	2 ms	15 m	1 m
Ground feature	0.001 pixel for PS	0.003 m	0.003 m
	1 pixel for DS	3 m	3 m
Earth tide		0.4 m	0 m
Earth rotation error		0.025 m	0 m
Ocean movement-related displacement		0.01 m	0 m
Atmospheric and hydrological loads	-	0.001 m	0 m
Electronic delay	1.2 Nanosecond	36 m	1 m
Ionospheric delay	5 TECU	1.29 m	1 m
Tropospheric delay	-	3 m	1 m
Total		39.139 m	2.000

The total positioning error of LT-1 is shown in Table II. The error before calibration is approximately 39.1 m. Most of the error sources could be calibrated except for the ground feature error. In this table, we do not consider that the external DSM error depends on the DSM types the users have. We only calculate the user-independent error sources.

B. Baseline Accuracy and Baseline Error Propagation

LT-1 uses total zero-Doppler steering in the satellite platform and zero-Doppler imaging technology in the ground segment [14]. Therefore, we did not consider the along-track baseline component. Baseline components in the zero-Doppler surface can be decomposed into two parts, i.e., components parallel and perpendicular to the LOS direction. The parallel and perpendicular baseline components in the zero-Doppler surface function differ. The error propagated from the parallel baseline to the height is [15]

$$\sigma h_{B_{\|}} = \frac{\sigma B_{\|}}{\lambda} \times h_{2\pi} \tag{4}$$

where $\sigma h_{B\|}$ is the elevation error induced by the parallel baseline error and $h_{2\pi}$ is the HoA. This is referred to as the wavelength magnitude effect, suggesting that a 23.5-cm parallel baseline error could increase or decrease the DSM height via the HoA. If the HoA is high, then the phase changes more quickly with ground undulation, and the phase

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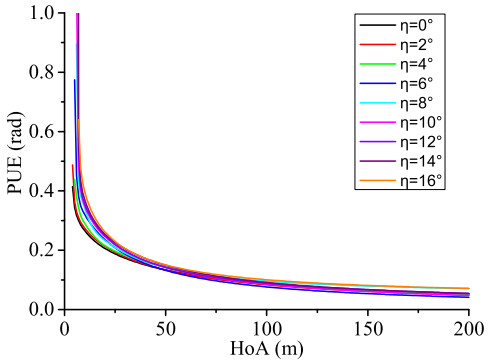


Fig. 11. Relationship between the HoA and the PUE.

unwrapping difficulty is high. The HoA can be expressed using the perpendicular baseline and looking angle as follows:

$$h_{2\pi} = \frac{\lambda R_A \sin(\theta - \eta)}{B_\perp} \tag{5}$$

where η is the local slope angle, $B_\perp = B\cos(\theta - \alpha)$, and α is the inclination angle. The HoA is inversely proportional to the perpendicular baseline. This is a conflict between the baseline length and baseline coherence. If the perpendicular baseline is large enough, the resulting interferometric phase aliasing could be great enough to introduce decoherence, especially in alpine regions where the local slope angle greatly fluctuates. We simulated different HoA values as a function of the slope angle. Then, we add different noise levels and unwrap the data to assess the PUE. Although the PUE is an integer multiple of 2π, it affects the final accuracy of DSM similar as gross error. The results are fit using the Levenberg–Marquardt + universal global optimization method and shown in Fig. 11 [6], [16]. The PUE was used for the first time as a quantitative value instead of an all-or-nothing influence. In operational processing, the result should remain as stable as possible. Therefore, the HoA value should not be less than 20 m. Otherwise, this may introduce at least a 0.6-m random elevation error even in flat terrain. We obtained statistics for LT-1, the perpendicular baselines are between 750.3 and 2156.5 m, and the mean value is 1791.5 m. The HoA values range from 19.9 to 147.5 m, and the mean value is 55.7 m. Typically, we can assume that both the inclination angle and slope angle are equal to zero. Then, in triangulation $\Delta S_A O P$, we can obtain the following equation by jointly considering (4) and (5)

$$\sigma h_{B_\parallel} = \left(H\cos\theta - \sqrt{r^2 - H^2\sin^2\theta}\right)\frac{\sin^2\theta}{\cos\theta}\frac{\sigma B}{B} \tag{6}$$

where H is the satellite altitude. The looking angle of the satellite ranges from 10° to 60°, which defines the lower and upper bounds of the satellite's observation capacity. The height error with respect to (6) is shown in Fig. 12. The baseline error greatly affects the final DSM quality if the baseline is very small. Fortunately, the looking angle difference within a whole scene is approximately 3°, resulting in a height error following a second-order polynomial in the case of 700 m at 1.2 cm (baseline length at error), and the residue is less than 0.8 m.

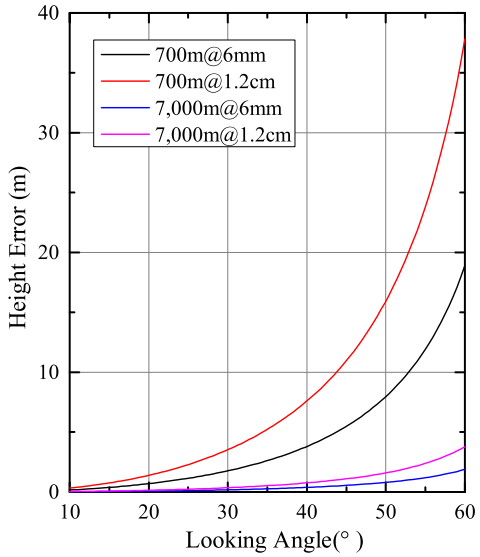


Fig. 12. Relationship between the baseline error and height error along the parallel baseline direction.

The perpendicular baseline accuracy exerts a minor effect. By considering (5), we can conclude the following:

$$\sigma h_{B_\perp} = -h\frac{\sigma B_\perp}{B_\perp} = -h\frac{\sigma B}{B} \tag{7}$$

where h is the height of the target to the Earth's reference ellipsoid surface. Therefore, when the perpendicular baseline is great enough, the perpendicular baseline-related height error can be suppressed to the centimeter level. In a specific image, the baseline is identical along the range direction, and the error related to the perpendicular baseline never induces a phase ramp along the range direction.

Baseline calibration can be conducted considering the differences in the propagation or errors between $B_\parallel$ and $B_\perp$. LT-1 supports InSAR DSM generation with looking angles ranging from 20° to 46°, and the baseline error can be suppressed within 1.2 cm. An obvious height error ramp is obtained along the range direction due to the parallel baseline error. After removing the $B_\parallel$ error, the $B_\perp$ error can be calculated with (7).

The contributions of the baseline error to the height error are shown in Table III. Given that the baseline length and looking angle change both spatially and temporally, we constructed a table of different terrain types and different HoA values. HoA of 96.2% HBF interferograms is smaller than 90 m, meaning that the elevation error is smaller than 5 m in all terrain types. However, the residue is expected to be less than 0.8 m after polynomial fitting.

C. Phase Accuracy and Characterization

The phase constant and random phase errors in the processing chain were considered. The phase constant error related to the instrument, i.e., the phase internal calibration error of the payload and the phase synchronization error between the two satellites, should be removed. The residuals should be less than 5°, and the related height error can be fit using a second-order polynomial. In this article, we mainly discuss the phase constant error related to the phase residue in the phase unwrapping procedure. Regardless of which phase unwrapping

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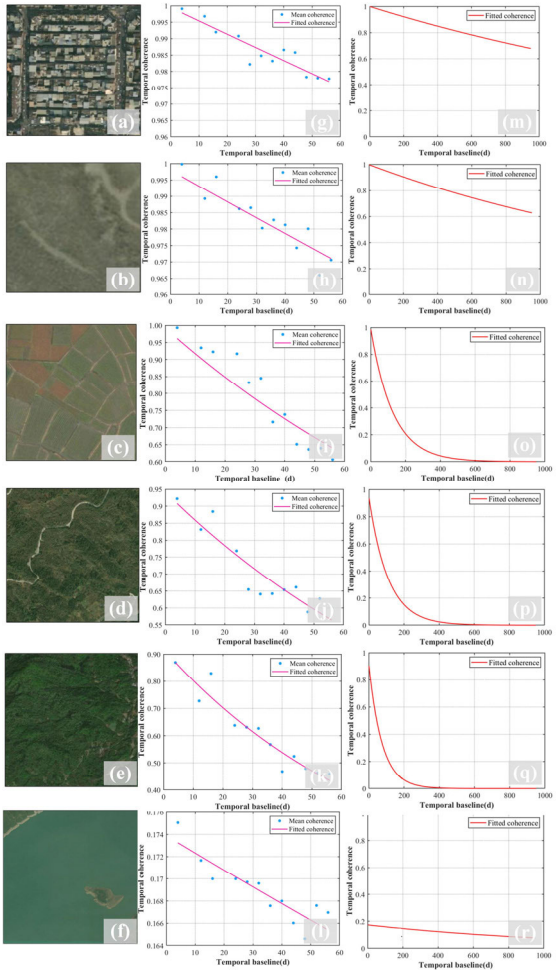


Fig. 13. (a)–(f) Optical images obtained from Gaofen-7 with land covers of building, bare land, water, farmland, shrub, and forest, respectively. (g)–(l) Temporal decorrelation with temporal baselines of 4, 12, 16, 24, 28, 32, 36, 40, 44, 48, and 52 days. (m)–(r) Temporal decorrelation fit using (13) with temporal baselines enlarged to 1000 days.

TABLE III

BASED ON THE PROPAGATION FROM THE BASELINE ERROR TO THE HEIGHT ERROR, THE PARALLEL BASELINE ERROR IS ASSUMED TO BE 1.2 cm

Slope / HoA	0	2	6	25
20	1.21	1.23	1.27	1.35
30	1.74	1.75	1.78	1.85
40	2.26	2.56	2.27	2.34
50	2.77	2.76	2.76	2.83
60	3.27	3.27	3.26	3.33
70	3.78	3.77	3.75	3.83
80	4.28	4.27	4.25	4.33
90	4.79	4.77	4.75	4.83
100	5.29	5.28	5.25	5.33
110	5.79	5.78	5.75	5.84
120	6.30	6.28	6.25	6.34
130	6.80	6.79	6.76	6.85
140	7.31	7.29	7.26	7.36
150	7.81	7.80	7.77	7.87

method is applied, the phase of a single pixel can always be expressed as

$$\varphi = \phi + 2k_1\pi + 2k_2\pi \tag{8}$$

where ϕ is the residual phase of the pixel, which can be calculated using the complex value; k_1 is the phase ambiguity of the starting point to be determined; and k_2 is the unwrapped phase ambiguity. k_1 and k_2 are all integers. k_1 is calculated from the modal number of thousands of observations using the radar geometry method [17]. Although the accuracy of radar geometry is low, it is enough to provide the integer number of k_1, and we do not need to consider the error of k_1. However, phase residue, which is affected by the random error, always exists. The relative height error can be expressed as

$$\sigma h_{\text{rel}} = h_{2\pi}\frac{\sigma\varphi}{2\pi}. \tag{9}$$

The relative height error is less than 2 m; therefore, the random phase error should be less than 36°, and the final coherence calculated using the Cramer–Rao criterion should be greater than 0.75 [18]. The coherence can be expressed using eight parts, as follows:

$$\gamma = \gamma_{\text{Bas}}\gamma_{\text{Tem}}\gamma_{\text{SNR}}\gamma_{\text{Vol}}\gamma_{\text{Amb}}\gamma_{\text{Qua}}\gamma_{\text{Dop}}\gamma_{\text{Pro}} \tag{10}$$

where the terms on the right-hand side include the baseline coherence (γ_{Bas}), temporal coherence (γ_{Tem}), signal-to-noise ratio (SNR) coherence (γ_{SNR}), volume coherence (γ_{Vol}), ambiguity coherence (γ_{Amb}), quantization coherence (γ_{Qua}), Doppler spectral coherence (γ_{Dop}), and processing coherence (γ_{Pro}). Two of the components including γ_{SNR} and γ_{Vol} are affected by the land cover types. We are not able to determine the land cover types, so during our discussion, we consider that γ_{SNR}, γ_{Amb}, and γ_{Qua} are determined by the payload, while γ_{Bas}, γ_{Tem}, γ_{Vol}, and γ_{Dop} are determined by the platform.

The noise-equal sigma zero (NESZ) is important for determining the quality of the payload. The NESZ of LT-1 is better than -28 dB for the interferometric beams. The ambiguities along the range and azimuth directions are less than -20 and -15 dB, respectively. Therefore, γ_{SNR} and γ_{Amb} are larger than 0.97 and 0.96, respectively. The quantization coherence is related to the signal loss in analog-to-digital conversion. We assessed coherence by calculating the coherence using interferograms obtained from different block adaptive quantization (BAQ) ratios. We suggest the use of 10:6 for SLC images to ensure that the coherence is greater than 0.94. LT-1 provides support for BAQs of 10:10, 10:6, 10:4, 10:3, and 10:2. We degraded the BAQ when possible after in-orbit testing, thus allowing more data to be transmitted from the satellite to the ground segment.

The platform provides fuel for satellite maneuvers, aiming at controlling the satellite position within a fixed orbital tube. If we use Stripmap 1 for deformation monitoring, the baseline coherence is greater than 0.97. The volume coherence is better than 0.86 if the HoA is larger than 35 m and the extinction coefficient is 1.0 dB/m, which is related to the type of the tree. The Doppler coherence is affected by the residue of the Doppler centroid and the Doppler bandwidth. The residue value was less than 10 Hz. Hence, the Doppler coherence is greater than 0.99. The SRO tube radius, which is controlled by the platform, is the determining factor of interferometry. This is manually controlled at eight-day intervals during the in-orbit testing period. The satellite supports self-maneuvering after in-orbit testing.

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TABLE IV
PROPAGATION FROM PHASE ERROR TO HEIGHT ERROR. THE HOA IS ASSUMED TO BE 55.7 m

Error sources	Typical value	Height error before filtering	Height error after filtering
Phase internal calibration error	5 degrees	0.77 m	0.77 m
Phase synchronization error	10.00 degrees	1.55 m	1.55 m
Baseline coherence	0.97	1.57 m	2.86 m
Temporal coherence	1.00	0.00 m	
SNR coherence	0.97	1.58 m	
Volume coherence	0.86	3.71 m	
Ambiguity coherence	0.96	1.83 m	
Quantization coherence	0.94	2.27 m	
Doppler spectral coherence	0.99	0.89 m	
Processing coherence	0.96	1.83 m	
Total	0.69	6.50 m	3.39 m

The processing parameters also cause decoherence. If we adopt the 16-point sinc interpolation method and ensure a coregistration accuracy better than 0.1 pixel, the processing coherence loss is less than 0.96 [18]. The random noise can be reduced in the processing chain by multilooking and filtering. We used an as large as possible multilooking factor, i.e., 8 × 8, to fulfill the needs of the 25-m grid size under the 1:50 000 measurement scale requirement. The smaller the DSM resolution, the lower the height accuracy. We applied the standard Goldstein filtering method for DSM production. The random noise could be reduced by more than 98% after applying the filtering method [19].

The contributions of the phase error to the height error are shown in Table IV. The phase internal calibration error and the phase synchronization error are independent and identically distributed. The decoherence error conforms to the multiplicative characteristic. The random error that is induced by the decoherence sources can be removed after filtering. The filtered coherence is expected to be greater than 0.9, and the corresponding phase error and height error values are 19.6° and 3.03 m, respectively. The final height error related to the interferometric phase is 3.54 m.

IV. SYSTEM PERFORMANCE FOR DEFORMATION MONITORING

There are mainly two kinds of deformation products for LT-1, they are deformation field generated using DInSAR and multitemporal deformation generated using MTInSAR. Although the kernels of the methods are different, they are both mainly affected by the baseline accuracy and the temporal decorrelation. Besides, MTInSAR was proposed to overcome the effects of baseline and decoherence. However, the linear modeling method introduces error for deformation velocity. We derived the propagation equation in this section.

A. Baseline Accuracy

In the deformation monitoring processing chain, we first conduct geometric calibration and no longer describe the positioning accuracy effect. The residual error is the geometric error related to the baseline and elevation.

The baseline error differs from that during the DSM generation task phase. When we use GNSS data for orbital determination, the absolute orbital accuracy reaches approximately 10 cm, which is lower than the relative orbital accuracy, as mentioned above. The baseline error can be increased to 14 cm given the following equation:

$$\sigma^2 B = \sigma^2 SA + \sigma^2 SB \tag{11}$$

where σSA is the location error of the LT-1A APC, and σSB is that of the LT-1B APC.

The parallel baseline-induced deformation error can be expressed as

$$\sigma d_{B\parallel} = \sigma B_{\parallel} = \sin(\theta - \alpha)\sigma B \tag{12}$$

where $\sigma d_{B\parallel}$ is the deformation error related to the parallel baseline. If the baseline error remains fixed, the deformation accuracy is affected only by the looking angle, inducing the error ranges from 2.39 to 5.04 cm. But a sinusoidal function fitting can be adopted to capture this effect.

The perpendicular baseline-induced deformation error is identical to that in (7). This error imposes a larger effect on the deformation accuracy than does the parallel baseline error because of the coefficient related to the Earth surface elevation. First, those regions suffer from layover and shadow. That decreases the visibility and sensibility of interferometry in deformation monitoring, especially when we want to conduct 3-D deformation decomposition. In LOS deformation monitoring, the layover changes the coherence as described in Section III-C. Meanwhile, the shadow is totally decoherent, and we should remove the regions. Second, in undulating areas where the elevation greatly changes, the related perpendicular deformation could exceed several meters, e.g., in the Everest Mountains. The DInSAR result should be better than 5 cm. We use the elevation deviation to define the four terrain types as flat ground, hilly region, mountains, and alpine regions. The baseline lengths corresponding to the terrain types are smaller than 224 m, 224 (included)–840 m, 840 (included)–1680 m, and larger than or equal to 1680 m if the baseline error is approximately 14 cm, respectively. This is the second conflict between the baseline error and baseline coherence. We prefer a short baseline to ensure interferometric coherence. Thus, the perpendicular baseline error could increase in mountainous and alpine regions. Therefore, baseline re-estimation in DInSAR processing procedures should always be applied. In addition, the elevation error affects the DInSAR results. If we use the SRTM data as elevation reference data, the height accuracy reaches approximately 16 m [5]; thus, the deformation error is approximately 3.2 mm if the baseline length is 700 m. If we decrease the baseline length to 100 m, the corresponding deformation error increases to 2.24 cm, which is acceptable during DInSAR processing. More precise DSM data are encouraged to be used for better DInSAR results. If we use Ziyuan-3 DSM with 5-m height accuracy or TanDEM-X DSM with 4-m height accuracy, the deformation error would decrease to 0.70 and 0.56 cm when using 100-m baseline length. We can even use Gaofen-7 DSM which fulfills the needs of 1:10 000 measurement scale. Table V shows the relationship between the baseline length and the deformation error. We calculated 60 921 interferograms and found that 90% of the perpendicular baseline values were less

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TABLE V
DEFORMATION ERROR INCLUDED THE BASELINE ERROR. THE BASELINE LENGTH IS 646 m. HOA IS ASSUMED TO BE 55.7 m

Terrain type	Perpendicular error related (cm)
Flat ground	0
Hilly regions	1.73
Mountains	6.50
Alpine regions	13.00

than 646 m. We used a perpendicular baseline of 646 m as follows. Interferometric beams allowed us to obtain the data using a looking angle between 20° and 46°. We used 33° as an example.

B. Temporal Decorrelation

The phase components are identical to those of the DSM generation task. In the deformation monitoring task, most decoherence is induced by temporal effects, given that we control the orbital tube to be as small as possible. The temporal coherence is a function related to the random location or physicochemical change in the object on the Earth's surface [20], which can be expressed as

$$\gamma_{\text{tem}} = \gamma_0 e^{-t/\tau} \tag{13}$$

where t is the temporal baseline, τ is the temporal decoherence rate, which is proportional to λ^2, and γ_0 is the initial coherence at the beginning of the temporal baseline, which can be expressed as follows:

$$\gamma_0 = \gamma_{\text{Bas}}\gamma_{\text{SNR}}\gamma_{\text{Vol}}\gamma_{\text{Amb}}\gamma_{\text{Qua}}\gamma_{\text{Dop}}\gamma_{\text{Pro}} \tag{14}$$

where the payload-related decoherence values are identical to those of the DSM generation task. We selected six kinds of land covers, i.e., building, bare land, water, farmland, shrub, and forest, to fit (13). The temporal baselines are 4, 12, 16, 24, 28, 32, 36, 40, 44, 48, and 52 days. If we enlarge the temporal baseline to 1000 days, the nonvegetated regions such as building areas and bare lands still own a higher coherence greater than 0.6, see Fig. 13. The short-vegetated areas are moderately correlated in the farmland and shrubs regions, and the coherence is higher than 0.3 within half a year. In the forest regions, LT-1 shows its extraordinary coherence which is higher than 0.3 when the temporal baseline is larger than two months. But the coherence of the water regions is always below 0.2 and even the temporal baseline is four days. There is only one way to improve the temporal correlation, decreasing the temporal baseline as small as possible. That depends on the orbital design procedure. Next generation of LT-1 is considering using a high resolution and wide-swath imaging technology, decreasing the minimum temporal baseline to less than two days.

C. MTInSAR Modeling Error

We used one of the typical, the PSInSAR technology to analyze the MTInSAR performance of LT-1. It is temporarily coherent point InSAR (TCPInSAR) method [21]. Compared with DInSAR and stacking, TCPInSAR removes the components of φ_{atm} and φ_{non} via spatiotemporal filtering, GNSS observations [12], or temporal modeling [21]. The deformation velocity and elevation error can be simultaneously modeled using more than 20 images along a single arc between two PS points as follows [22]:

$$\Delta\varphi_{(x,y)k} = -\frac{4\pi}{\lambda}\Delta v_{(x,y)}T_k - \frac{4\pi B_{\perp k}}{\lambda R \sin\theta}\Delta\varepsilon_{(x,y)} + w\varphi_{(x,y)k} \tag{15}$$

where x and y are the PS indices along the kth arc, $\Delta v_{(x,y)}$ is the relative deformation velocity along the kth arc, $\Delta\varepsilon_{(x,y)}$ is the relative elevation error, and $w\varphi_{(x,y)k}$ is the phase residue of the arc. If we obtain more than two interferograms, we can apply the least-square estimation (LSE) method to calculate the observations with (15). However, more than 20 images are suggested to provide tens of interferograms and improve the LSE robustness. DInSAR and stacking failed to calculate the elevation error, and the accuracies of them are both lower than PSInSAR theoretically. The relative deformation along arcs can be converted into the absolute deformation of points using the overall LSE. Similarly, we can obtain the deformation velocity accuracy as follows:

$$\begin{aligned}\sigma_V &= \frac{\lambda}{4\pi}\sigma\varphi_v \\ &= \frac{\lambda}{4\pi}\sqrt{\frac{1}{N}\sum_{i=1}^{N}(\varphi_i - \varphi_{\text{flt}} - \varphi_{\text{hgt}} - \varphi_{\text{atm}} - \varphi_{\text{noi}} - \varphi_{\text{non}})^2}.\end{aligned} \tag{16}$$

The theoretical accuracy improvement in PSInSAR deformation velocity is obvious if the components are correctly obtained. The main error sources of (16) are flattening phase error, height error, atmospheric error, noise error, and nonlinear deformation error. The error propagation coefficients of the four components are $1/\sqrt{N}$ of those in DInSAR, where N is the number of interferograms. If we use more than 30 images, the components of the flatting phase residues are 4.79% of those in (12). The deformation error related to the parallel baseline will be smaller than 1.76 mm. The height error is modeled in (15), but we still need a more precise DSM which can provide better results because the baseline error is coupling with height error in the interferometric phase. If we use the baseline length of 100 m, the corresponding deformation errors caused by the height errors of SRTM, Ziyuan-3, and TanDEM-X are 0.57, 0.26, and 0.20 mm, respectively. Atmospheric error is vital, and the ionospheric and tropospheric phase advances are 0.97 and 2.7 m maximumly for a single image, but we have been provided a specially developed GACOS module to remove atmospheric phase screen for differentiation [23]. For the ionosphere effect, we have adopted the range split-spectrum method given that the bandwidth of LT-1 is around 80 MHz. The atmospheric residue of an interferogram is about several centimeters for DInSAR, and a maximum of 3 mm for MTInSAR. Meanwhile, the spatiotemporal filtering method is applied dealing with the phase residues, which decrease the noise error effect of MTInSAR further. The last component of (15) is not stochastic. If the quadratic or cubic term related to the temporal baseline is modeled instead of simply the linear deformation

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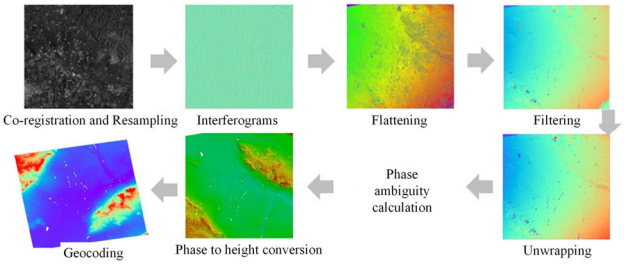


Fig. 14. Standard processing procedures for InSAR DSM generation.

term, the nonlinear deformation estimation accuracy could increase. However, in operational product generation applications, we should still adopt a linear model to ensure standard and repeatable processing procedures. Compared with DInSAR and stacking, TCPInSAR provides high accuracy but a low observation density. This technology is useful for detecting the deformation of artificial buildings. If the observation accuracy is better than 10 mm/year, the expected phase dispersion of N images remains within 0.53 rad, which is approximately 30.6° in a year. We observe China once every 28 days; thus, the single phase error should be within $30.6 \times \sqrt{13} = 110°$, which is larger than the general phase residue of 20°.

V. First Results and Analysis

The in-orbit LT-1 test lasted for approximately one year. We have established the calibration fields and CRs shown in Fig. 10. All the images send to us will be calibrated and updated within 24 h. Then we need to check for the radio frequency interference and remove the effects [24], [25]. After that, we obtained preliminary DSM and deformation results.

We use the standard processing procedures of InSAR technology, which are shown in Fig. 14. First, the images were coregistered using the bistatic imaging mode, which is different from the traditional range–Doppler (R–D) model. The slant range of the secondary satellite is affected by both the LT-1A and LT-1B satellites, i.e.,

$$\left|\overrightarrow{S_A P}\right| + \left|\overrightarrow{S_B P}\right| = 2R_B. \tag{17}$$

The Doppler equation is also converted to

$$\frac{\overrightarrow{V_A} \cdot \overrightarrow{\mathrm{PS}_A}}{\lambda \left|\overrightarrow{\mathrm{PS}_A}\right|} + \frac{\overrightarrow{V_B} \cdot \overrightarrow{\mathrm{PS}_B}}{\lambda \left|\overrightarrow{\mathrm{PS}_B}\right|} = f_B \tag{18}$$

where f_B is the Doppler centroid frequency of the secondary satellite image. Primary and secondary satellites are exchanged considering the satellite imaging plans and the satellite fuels. The image pairs are distributed to the user using the prefix of LT1AB, which indicates that the primary image is LT-1 A, while LT1BA, which indicates that the primary image is LT-1B. We use the path integral phase unwrapping method [16]. The L2-norm phase unwrapping methods such as minimum cost flow (MCF) should be avoided because they change the phase residues after unwrapping, thus misleading the height accuracy. After phase unwrapping, we estimate the phase ambiguities using interferometric radargrammetry [26]. The ambiguities of the DSM data covering Shangqiu and Suqian are all zero. Finally, we remove the systematic height error

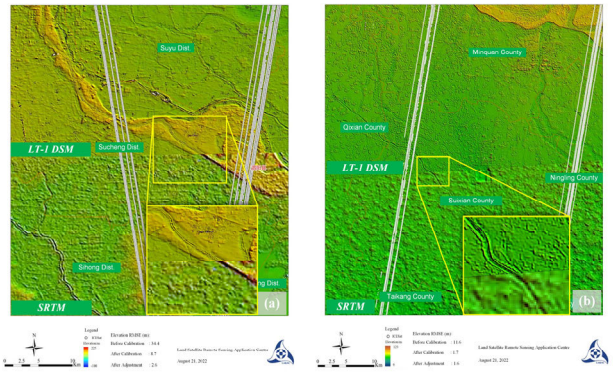


Fig. 15. DSMs were generated using LT-1 in the upper part of the images, and by the SRTM C-band in the lower part of the images. (a) Suqian city in Jiangsu Province. (b) Shangqiu city in Henan Province. The white points indicate the ICESat data. Yellow rectangles are the partially enlarged detail comparisons between the LT-1 DSM and SRTM.

introduced by the residues related to the baseline, interferometric phase, slant range, azimuth timing, and so on using the Ice, Cloud, and Land Elevation Satellite (ICESat) data. If no ICESat data were available, then we should use the DSM data over a large continuous land to conduct LSE to ensure the internal consistency. If we could collect data all over the world, we should use LSE over continents. During this processing, the points along the coastlines are deemed as control points and their elevation values should be set to 0 [5].

Two DSM products generated using LT-1 are shown in Fig. 15. Two calibration fields are used to calibrate the baseline error. The figure reveals that the Earth surface undulations in the LT-1 and SRTM DSMs are similar. The original range and azimuth pixel spacing of LT-1 are 1.7 and 1.8 m, respectively. We produced the DSM data with a grid size of 12 m. Roofs of Suqian were observed in the LT-1 DSM data, but no buildings could be recognized in the SRTM data due to the 30-m intrinsic resolution of the product. However, in Fig. 15(b), the watercourse is consistent between LT-1 and SRTM. The 90% circular error (CE90) of SRTM data is better than 13 m [5], while that of LT-1 data is better than 4 m which is assessed using CRs. There might be at most a 17-m relative error which is barely visible compared with the 30-m resolution of SRTM data.

The white points are extracted from the ICESat altimeter payload. The height accuracy of ICESat-1 was better than 15 cm [27]. ICESat-1 data were used to remove the phase ramp induced by interferometric parameter error residues, including the range, azimuth timing, baseline, phase, and PAC position, and provide an accuracy reference for the elevation. Moreover, we cleaned the original ICESat-2 original data to make the system more compatible. Statistics indicate that the RMSE before calibration is 11.6 m, and that after calibration is 1.7 m. After processing, the height accuracy could increase to 1.6 m. In Suqian city of Jiangsu Province, as shown in Fig. 15(b), the corresponding accuracy values before calibration, after calibration, and after ICESat correction are 34.4, 8.7, and 2.6 m, respectively.

After calibration, we produced DSM data covering an area of 330 000 km^2 in China. Some typical DSMs over here and

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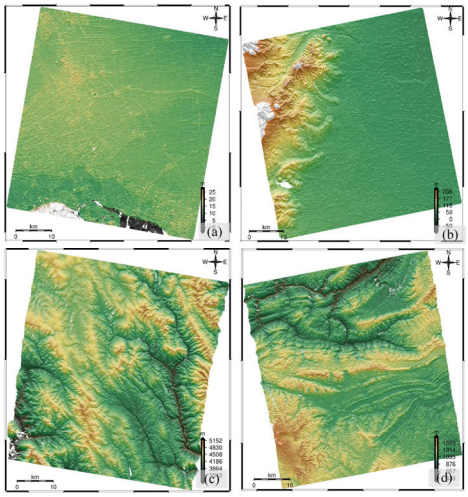


Fig. 16. DSMs were obtained in flat regions, hills, and mountainous cloudy-and-rainy areas located at Jiangsu, Hebei, Sichuan, and Guizhou.

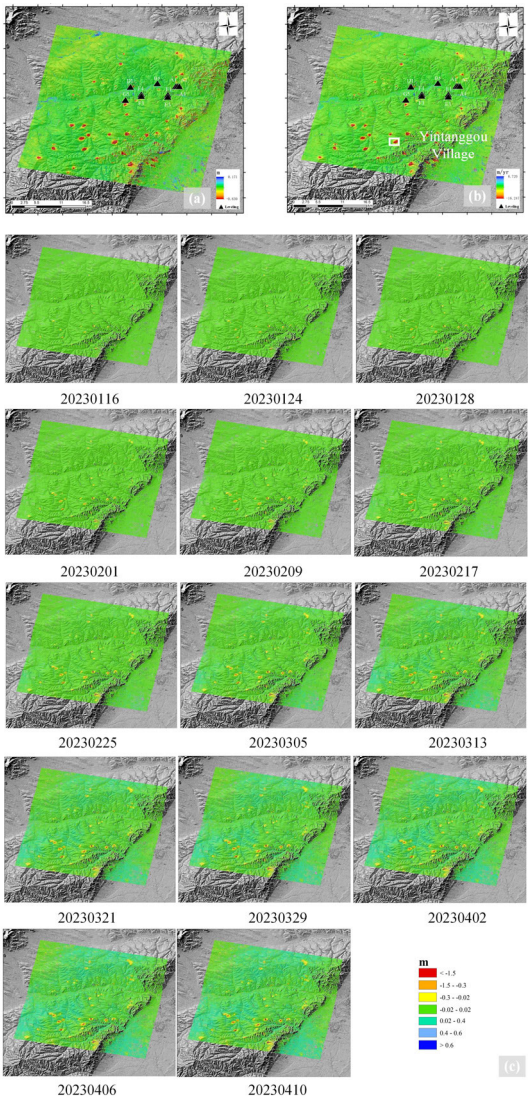


Fig. 17. Deformation products generated using LT-1. (a) Generated using DInSAR. (b) Deformation velocity generated using TCPInSAR. (c) Time-series results.

other in-orbit test regions are shown in Fig. 16. In the flat region of Jiangsu, the DSM accuracy values before and after

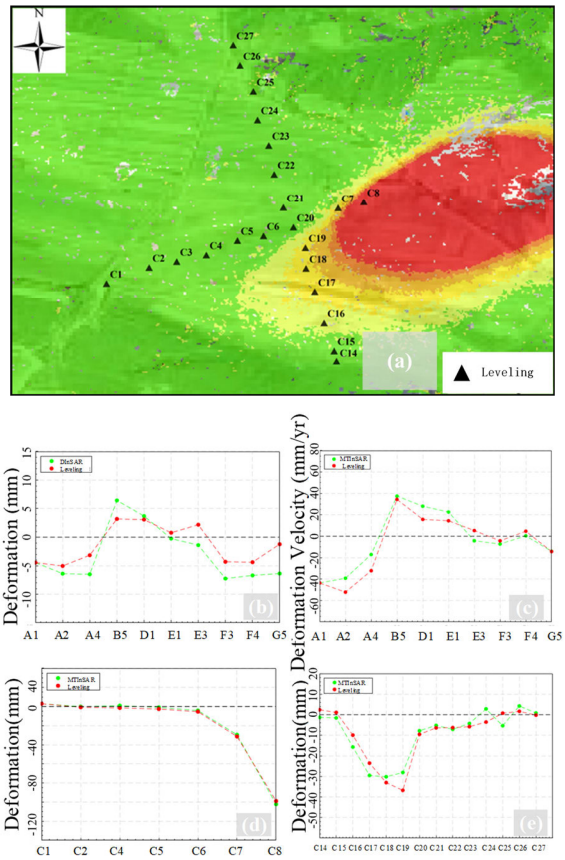


Fig. 18. Comparison using leveling data. (a) Leveling data distributed along and across Yintanggou Village and are used for time-series comparison. (b) Comparison between DInSAR and the leveling data in Fig. 17(a). (c) Comparison between TCPInSAR deformation velocity and the leveling data in Fig. 17(b). (d) and (e) Comparisons between TCPInSAR and leveling in the along and across mining directions, respectively.

ICESat correction are 1.2 and 1.0 m, respectively; those of the hills in Hebei are 0.9 and 0.7 m, respectively; and those of the mountainous cloudy and rainy regions in Xizang and Guizhou are 5.9 and 6.7 m, respectively. The white holes in the figures are those regions with coherence lower than 0.9.

To ensure the accuracy and consistency of the DSM data, we collected the corresponding data for at least three times. Before September 25, 2022, we measured the regions of interest using descending and ascending geometries with short baselines, which ensured that the HoA was approximately 60 m. Thereafter, we spent less than ten days converting the formation into the long-baseline configuration, thus reducing the HoA to approximately 35 m. All the regions were covered during the fourth measurement. Unfortunately, we did not have enough time to observe the region more than four times. The satellite was reverted to PMF at the end of November 2022. The remaining lifetime of the satellite will be spent on the deformation monitoring task, which is the main task of these satellites. Moreover, we do not have enough fuel to change to HBF and change back. We leave the DSM generation tasks to other satellites such as Ziyuan-3 and Gaofen-7.

Fig. 17 shows the results of DInSAR and MTInSAR for Datong city in Shanxi Province. The DInSAR results are provided using the primary and secondary images taken between

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December 15, 2022 and January 16, 2023. The temporal baseline is 33 days, and the perpendicular baseline is 214 m. The TCPInSAR results are shown with the deformation velocity and time-series in Fig. 17(b) and (c). They are calculated using 14 images taken between January 16, 2023 and April 04, 2023. Although the deformation trend is similar between DInSAR and TCPInSAR, the subsidence funnels are different. There are 46 subsidence funnels detected in DInSAR, but only 41 in TCPInSAR, meaning that at least five regions are dominated by nonlinear deformation.

The leveling data were collected on the same day that the satellite flew over Datong. They are marked in Fig. 17(a) and (b) and Fig. 18(a) using black triangles. The RMSE of DInSAR is 2.7 mm; see Fig. 18(b). The deformation velocity is assessed using the same points of DInSAR, and the RMSE is 8.6 mm/year; see Fig. 18(c). To further validate the accuracy of TCPInSAR, we use extra leveling data taken along and across the mining directions; see Fig. 18(a). The RMSE is 3.7 mm in Fig. 18(d) and (e) when assessing using those all the leveling points.

VI. Conclusion

In this article, we provide an overview of the design and realization of the LT-1 SAR satellite constellation and preliminary results. The satellite uses HBF to provide DSM data during the in-orbit test period. Then, PMF is adopted to provide deformation data. Deformation monitoring is the primary task. LT-1 uses an orbital tube radius of 350 m and an orbital revisit time of four days to ensure the quality of the deformation monitoring. We also provide preliminary DSM results covering the cities of Suqian and Shangqiu cities, as well as deformation data covering Datong city.

The satellites are now used to conduct deformation monitoring over the HIMSGA. The satellite constellation traverses the HIMSGA once 28 days due to the limitation of the data down transfer rate of 2×450 Mb/s. DInSAR and MTInSAR are the main technologies to produce deformation information. We are now considering to separate stacking from MTInSAR to provide more product options. As we collect more and more images, more MTInSAR methods including PSI, SBAS, SqueeSAR, etc. will be assessed soon. We hope that the satellite will provide enough information to locate potential geohazards and protect the property and lives of people at risk.

References

[1] R. F. Hanssen, "Atmospheric hetero geneities from ERS TanDEM SAR interferometry and sea surface images," in *Proc. CEOS SAR Workshop*, Norrdwijk, The Netherlands, 1998, pp. 989–992.

[2] G. Krieger et al., "TanDEM-X: A satellite formation for high-resolution SAR interferometry," *IEEE Trans. Geosci. Remote Sens.*, vol. 45, no. 11, pp. 3317–3341, Nov. 2007.

[3] A. Moreira et al., "Tandem-L: A highly innovative bistatic SAR mission for global observation of dynamic processes on the Earth's surface," *IEEE Geosci. Remote Sens. Mag.*, vol. 3, no. 2, pp. 8–23, Jun. 2015.

[4] J. F. L. Dekker et al., "The harmony mission: Applications and preliminary performance," Presented at the 6th Workshop Adv. RF Sensors Remote Sens. Instrum. (ARSI), Noordwijk, Netherlands, Jan. 2019.

[5] T. G. Farr et al., "The shuttle radar topography mission," *Rev. Geophys.*, vol. 45, no. 2, pp. 361–393, May 2007.

[6] S. Li, S. Zhang, T. Li, Y. Gao, Q. Chen, and X. Zhang, "Modeling the optimal baseline for a spaceborne bistatic SAR system to generate DEMs," *ISPRS Int. J. Geoinf.*, vol. 9, no. 2, p. 108, Feb. 2020.

[7] O. Montenbruck, M. Wermuth, and R. Kahle, "GPS based relative navigation for the TanDEM-X mission-first flight results," *Navigation*, vol. 58, no. 4, pp. 293–304, Dec. 2011.

[8] W. Yu et al., "The LuTan-1 SAR antenna system," presented at the EUSAR, Leipzig, Germany, Jul. 2022.

[9] G. Jin et al., "An advanced phase synchronization scheme for LT-1," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 3, pp. 1735–1746, Mar. 2020.

[10] T. Li, X. Tang, X. Gao, Q. Chen, and X. Zhang, "Analysis and outlook of the operational topographic surveying and mapping capacity of the SAR satellites," *Acta Geodaetica et Cartographica Sinica*, vol. 50, pp. 891–904, Feb. 2021.

[11] C. Gisinger et al., "Precise three-dimensional stereo localization of corner reflectors and persistent scatterers with TerraSAR-X," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 4, pp. 1782–1803, Apr. 2015.

[12] C. Yu, N. T. Penna, and Z. Li, "Generation of real-time mode high-resolution water vapor fields from GPS observations," *J. Geophys. Res., Atmos.*, vol. 122, no. 3, pp. 2008–2025, Feb. 2017.

[13] C. Yu, N. T. Penna, and Z. Li, "Ocean tide loading effects on InSAR observations over wide regions," *Geophys. Res. Lett.*, vol. 47, no. 15, pp. 1–10, Aug. 2020.

[14] Y. Liu, "Key techniques of signal processing for high-resolution and widh-swath distributed satellite SAR/InSAR," Xidian Univ., 2013.

[15] J. Hueso González et al., "Bistatic system and baseline calibration in TanDEM-X to ensure the global digital elevation model quality," *ISPRS J. Photogramm. Remote Sens.*, vol. 73, pp. 3–11, Sep. 2012.

[16] Y. Gao, S. Zhang, T. Li, Q. Chen, X. Zhang, and S. Li, "Refined two-stage programming approach of phase unwrapping for multi-baseline SAR interferograms using the unscented Kalman filter," *Remote Sens.*, vol. 11, no. 2, pp. 199–217, Jan. 2019.

[17] M. Eineder, C. Minet, P. Steigenberger, X. Cong, and T. Fritz, "Imaging geodesy—Toward centimeter-level ranging accuracy with TerraSAR-X," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 2, pp. 661–671, Feb. 2011.

[18] R. F. Hanssen, *Radar Interferometry: Data Interpretation and Error Analysis*. Alphen aan den Rijn, The Netherlands: Kuwer Academic, 2001.

[19] Y. Gao, S. Zhang, T. Li, L. Guo, Q. Chen, and S. Li, "A novel two-step noise reduction approach for interferometric phase images," *Opt. Lasers Eng.*, vol. 121, pp. 1–10, Oct. 2019.

[20] Y. Morishita and R. F. Hanssen, "Temporal decorrelation in L-, C-, and X-band satellite radar interferometry for pasture on drained peat soils," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 2, pp. 1096–1104, Feb. 2015.

[21] L. Zhang, "Temporarily coherent point SAR interferometry," Ph.D. thesis, Dept. Doctor Philosiphy, The Hong Kong Polytechnic University, Hong Kong, 2012.

[22] L. Zhang, X. Ding, and Z. Lu, "Modeling PSInSAR time series without phase unwrapping," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 1, pp. 547–556, Jan. 2011.

[23] R. Xiao, C. Yu, Z. Li, and X. He, "Statistical assessment metrics for InSAR atmospheric correction: Applications to generic atmospheric correction online service for InSAR (GACOS) in eastern China," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 96, Apr. 2021, Art. no. 102289.

[24] J. Chen et al., "Characterization and mitigation of radio frequency interference signatures in L-band LuTan-1 InSAR system: First results and assessment," *IEEE Trans. Radar Syst.*, vol. 2, no. 5, pp. 404–420, May 2024.

[25] Y. Cai et al., "First demonstration of RFI mitigation in the phase synchronization of LT-1 bistatic SAR," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, Sep. 2023, Art. no. 5217319.

[26] R. Bamler, "Interferometric stereo radargrammetry: Absolute height determination from ERS-ENVISAT interferograms," presented at the IGARSS 2000, vol. 2, Honolulu, HI, USA, 2000.

[27] C. A. Shuman et al., "ICESat Antarctic elevation data: Preliminary precision and accuracy assessment," *Geophys. Res. Lett.*, vol. 33, no. 7, Apr. 2006, Art. no. L07501.

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